Assignment

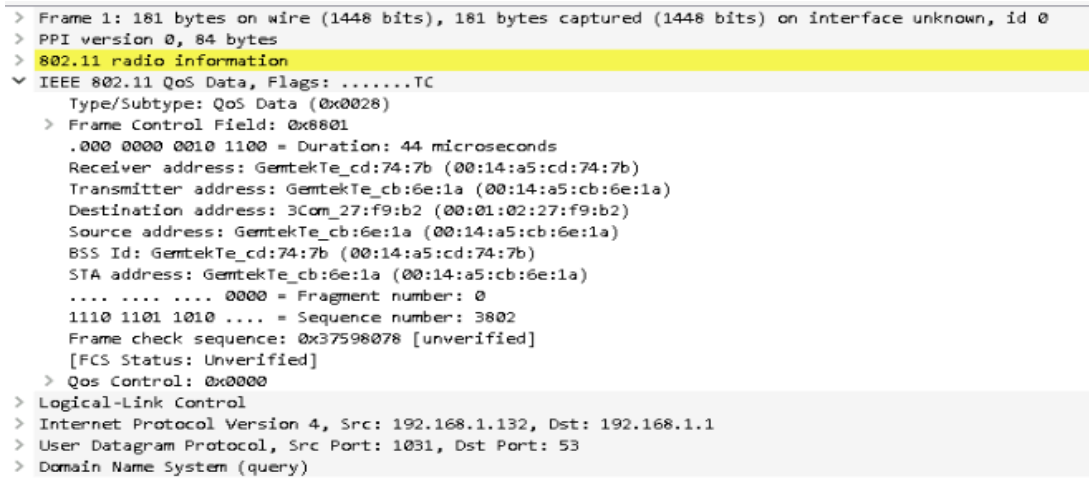
Name: Jonathan Hernandez (2651994)

Lab Screenshots and Narrative

1. Insert and title all lab screenshot(s) in the order they were taken.

Section 1, Part 1

In the Wireshark application, I successfully reviewed the packet details pane for the first packet. I started by collapsing the Frame Header detail by clicking the down arrow at the beginning of the Frame 1 line. Next, I expanded the Flags line to display the Flags detail, where I observed that the alignment was set to 0, indicating that the next byte after the field contains the next field. The header length was 84 octets, referring to the length of the PPI header only, and a Data Link Type (DLT) of 105 indicated that data was transferred over an 802.11n wireless network. I then collapsed the Flags line and expanded the 802.11-Common line to view fields common to all 802.11 wireless protocols. This included specific information about radio frequencies and channels, with fields indicating that the maximum rate of transmission was 300 Mbps (rate: 300.0 Mbps). After collapsing the 802.11-Common line, I expanded the 802.11n MAC+PHY line to display additional details related to MAC addressing and physical attributes. Using the scrollbar as necessary, I noticed a large amount of data about the 802.11n connection and the wireless environment, including signal strengths, noise ratios, and other antenna information. Finally, I collapsed the PPI version 0 line and expanded the IEEE 802.11 QoS Data, Flags line. Here, Wireshark displayed information about the transmitters and receivers, enabling the identification of Media Access Control (MAC) addresses matching each transmitter and receiver. A screenshot of the completed steps has been captured and is provided.



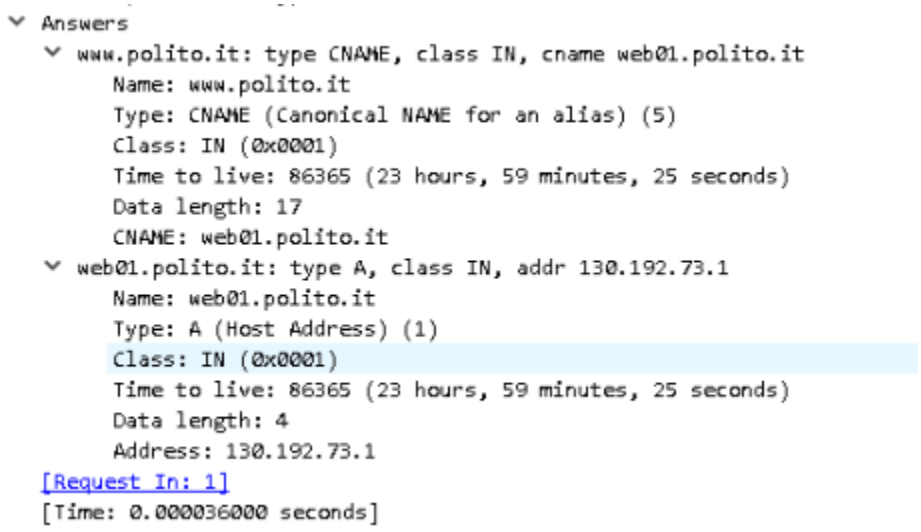
I successfully completed a detailed review of various lines and fields within the packet details pane. I began by expanding the QoS Control line and studied the fields and their values, noting their existence as part of the lab's scope without delving into their specific meanings or interactions. Next, I collapsed the Logical-Link Control line.

I then expanded the Internet Protocol Version 4 line and familiarized myself with the data available, expanding each heading within this section to thoroughly understand the available data, before collapsing the Internet Protocol Version 4 line. Following this, I expanded the User Datagram Protocol line and reviewed the data available, subsequently collapsing it after familiarization.

I expanded the Domain Name System (query) line to view the fields related to an Internet query. Within this section, I expanded the Queries line and noticed data indicating that someone attempted to access the www.polito.it website.

I then moved on to explore packets 2 and 3. In the packet list pane, I clicked on packet 2 and reviewed the packet details pane, noting that packet 2 is a wireless command and control packet acknowledging receipt of Packet 1.

I continued to explore the packet details pane by expanding each heading in the Answers section and familiarizing myself with the data available. These fields detailed the response to the DNS query explored in Packet 1. The data shown in these fields included the IP address for polito.it (130.192.73.1) and other DNS information, such as a DNS time to live (TTL) of 23 hours, 59 minutes, and 25 seconds, indicating the duration before the DNS cache for this entry must be refreshed.



Section 1, Part 2

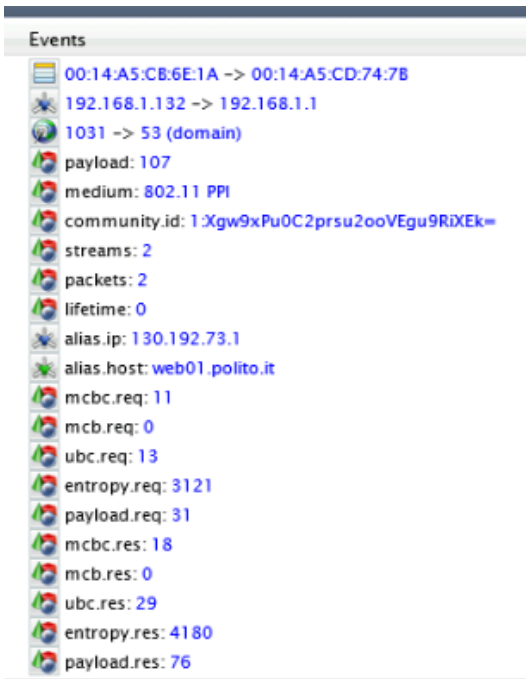
From the RSA NetWitness Investigator menu bar, I clicked on Collection and selected New Local Collection to create a new local collection. In the Collection Name box, I typed Jonathan_DemoCapture_S1, replacing "yourname" with my own name, and clicked OK to name the new collection.

Creating a new local collection within NetWitness Investigator provides a place to put the packets from the DemoCapture file. This new collection appeared in the left pane, the Collection pane of NetWitness Investigator. In the Collection pane, I double-clicked the Jonathan_DemoCapture_S1 collection I just created to activate it and change the status to Ready.

In the Open dialog box, I clicked the Desktop button, then double-clicked the DemoCapture4NI file that I saved in Part 1 to begin the import process. The Collection pane displayed a progress report while the import was underway. When the import was finished, the new collection again displayed a status of Ready.

Under the Service Type category, I clicked the green (1) link next to the DNS link to drill down and get further information about the DNS request. The (1) link following the DNS label indicated that there was only one DNS request in this packet capture file.

After comparing the information provided by NetWitness to the screen capture of the query name (www.polito.it), the source IP address, and the destination IP address in Wireshark, I can confirm that both tools indeed show the same IP and domain information. They accurately display details such as the source and destination addresses, time to live (TTL), and many other pertinent details. This consistency between the two tools reinforces the accuracy and reliability of the network analysis.



After comparing the information provided by NetWitness to the screen capture of the query name (www.polito.it), the source IP address, and the destination IP address in Wireshark, I can confirm that both tools indeed show the same IP and domain information. They accurately display details such as the source and destination addresses, time to live (TTL), and many other pertinent details. This consistency between the two tools reinforces the accuracy and reliability of the network analysis.

Upon comparing the information provided by NetWitness to the screen capture made of the detail found in the IEEE 802.11 QoS Data fields in Wireshark, I observed that while both tools display Ethernet source information, Wireshark provides more detailed information. Specifically, Wireshark includes additional details such as MAC Address, Manufacturer MAC Address, Fragments, and Sequences. This additional information enhances the depth of analysis and provides further insight into the network traffic.



Under the Destination City category, I clicked the green (1) link next to the Turin link to reveal additional details from this report.

From the data presented, I determined that the transaction originated in Turin, Italy, and was an HTTP get request in which a website was retrieved. NetWitness has conducted a comprehensive analysis of the higher-level transaction without revealing the lower-level frame or packet detail to the user.

It's important to note that while the Top-Level Domain (TLD) ".it" indicates Italy, there's no guarantee that the website is physically located in Italy. The presence of a domain name registered with the .it TLD only signifies registration with the appropriate registrar. To determine the actual physical location of the server hosting the website, geolocation technology such as IP-geolocation or triangulation using PINGs would be necessary.

Section 2, Part 1

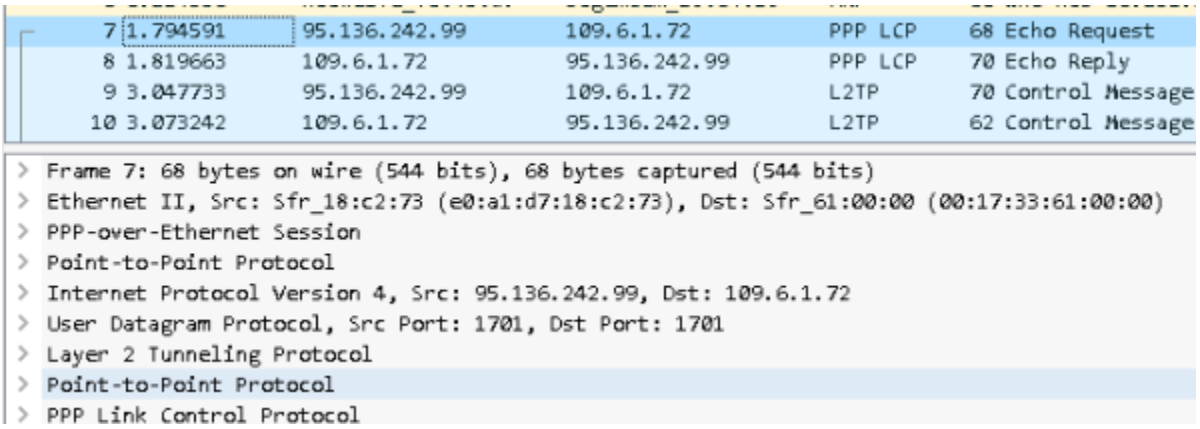
In Section 2 of the lab, I will apply the knowledge gained from Section 1 with less guidance and different deliverables, as well as explore expanded tasks and alternative methods. This section involves examining a different static PCAP file to further enhance my understanding of network analysis and investigation techniques.

In the Wireshark application, I opened the HotspotCapture.pcapng file located at C:\PCAPs\HotspotCapture.pcapng.

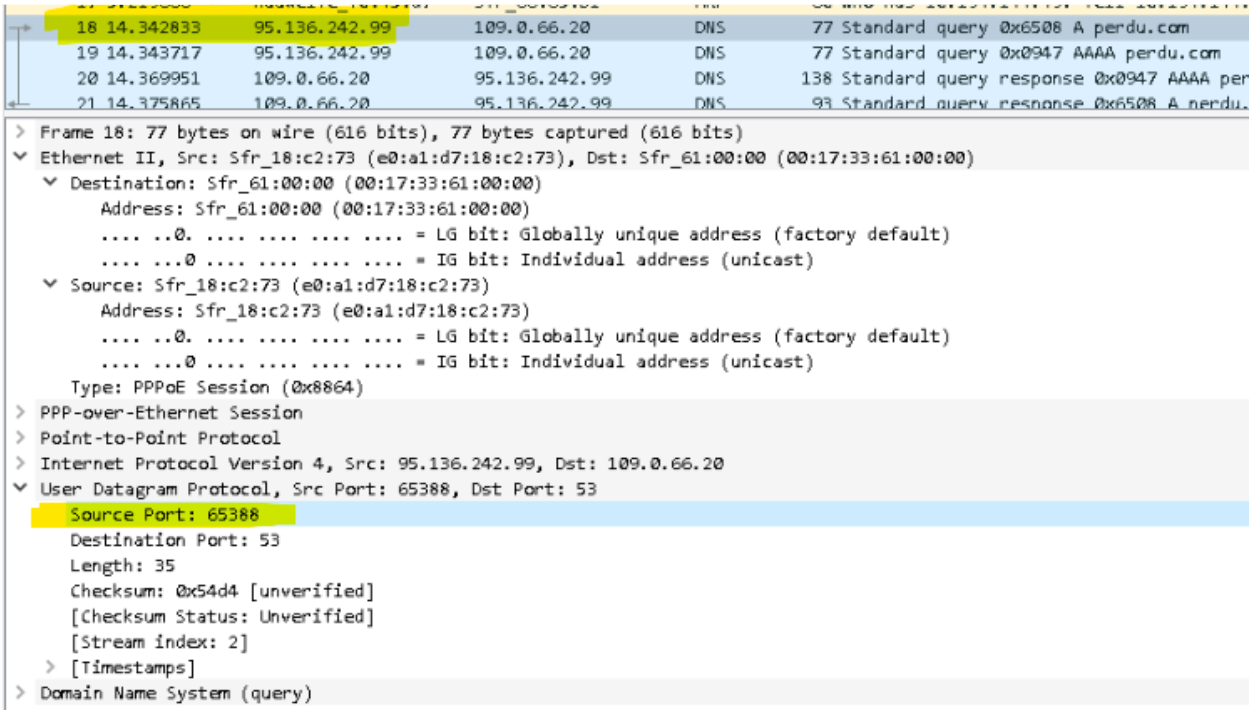
Next, I selected Packet 7 in the packet list pane.

To focus on packets 7-9 in the packet list pane, I adjusted the top border of the packet details pane accordingly.

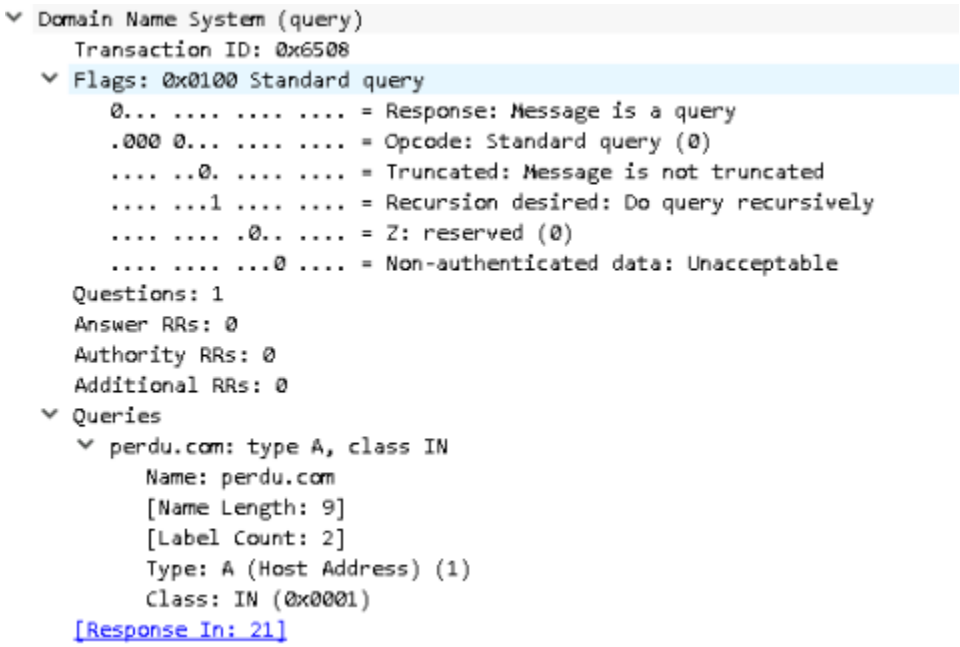
In the packet details pane, I reviewed the protocol list to identify the different protocols used in Packet 7.



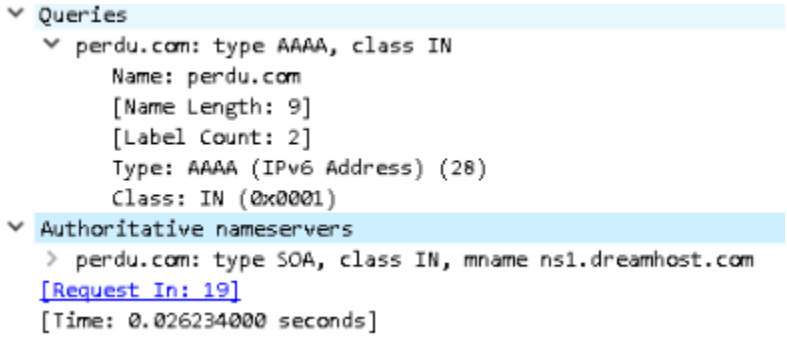
I'll capture the screen showing the source port number used in Packet 18 and also expand the Domain Name System (query) line, followed by expanding the Queries line to identify the domain queried. We are able to see that the Source Port is 65388



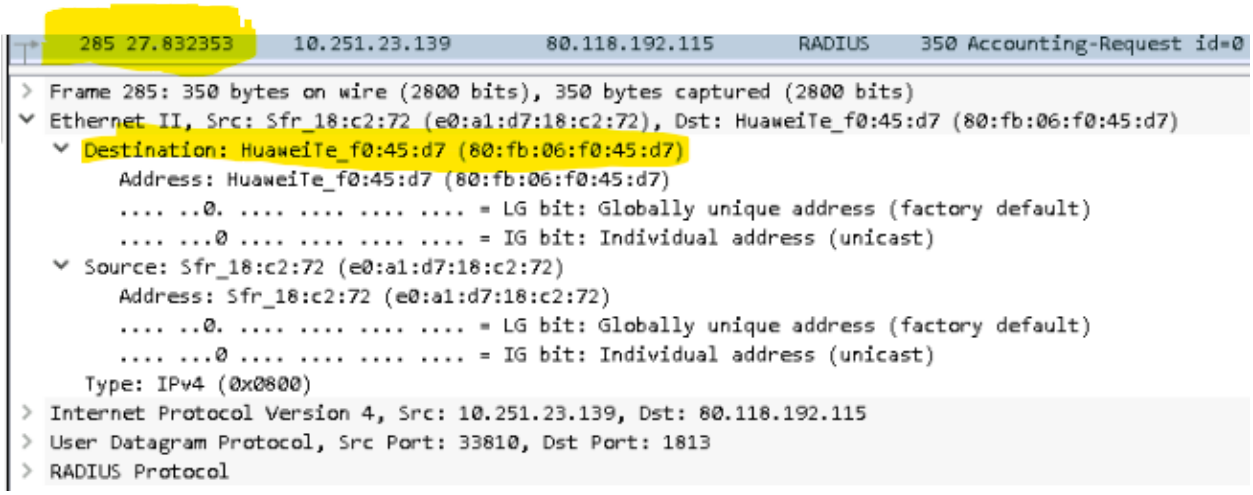
To identify the related packet using the information in Packet 18, I'll look for the packet that corresponds to the DNS query response. Once found, I'll select that packet in the packet list pane. Then, in the packet details pane, I'll expand the Answers line to view the details of the DNS query response, including the IP address for perdu.com and other DNS information such as the DNS time to live (TTL).



I have successfully selected Packet 20 in the packet list pane and expanded the Authoritative nameservers line to identify the details about the DNS records for the domain queried. In the captured screen, the answer to the Authoritative nameservers line states "perdu.com: type SOA, class IN, name ns1.dreamhost.com."



In the packet list pane, I've located Packet 285 and identified the destination MAC address. This address is vital for network communication, ensuring data transmission to the intended recipient. Wireshark provides insights into transmitters and receivers, aiding network troubleshooting. MAC addresses appear in hexadecimal and shorthand formats, translating to manufacturer names. The destination MAC address in Packet 285 is 80:fb:06:f0:45. This address is crucial for successful packet delivery.

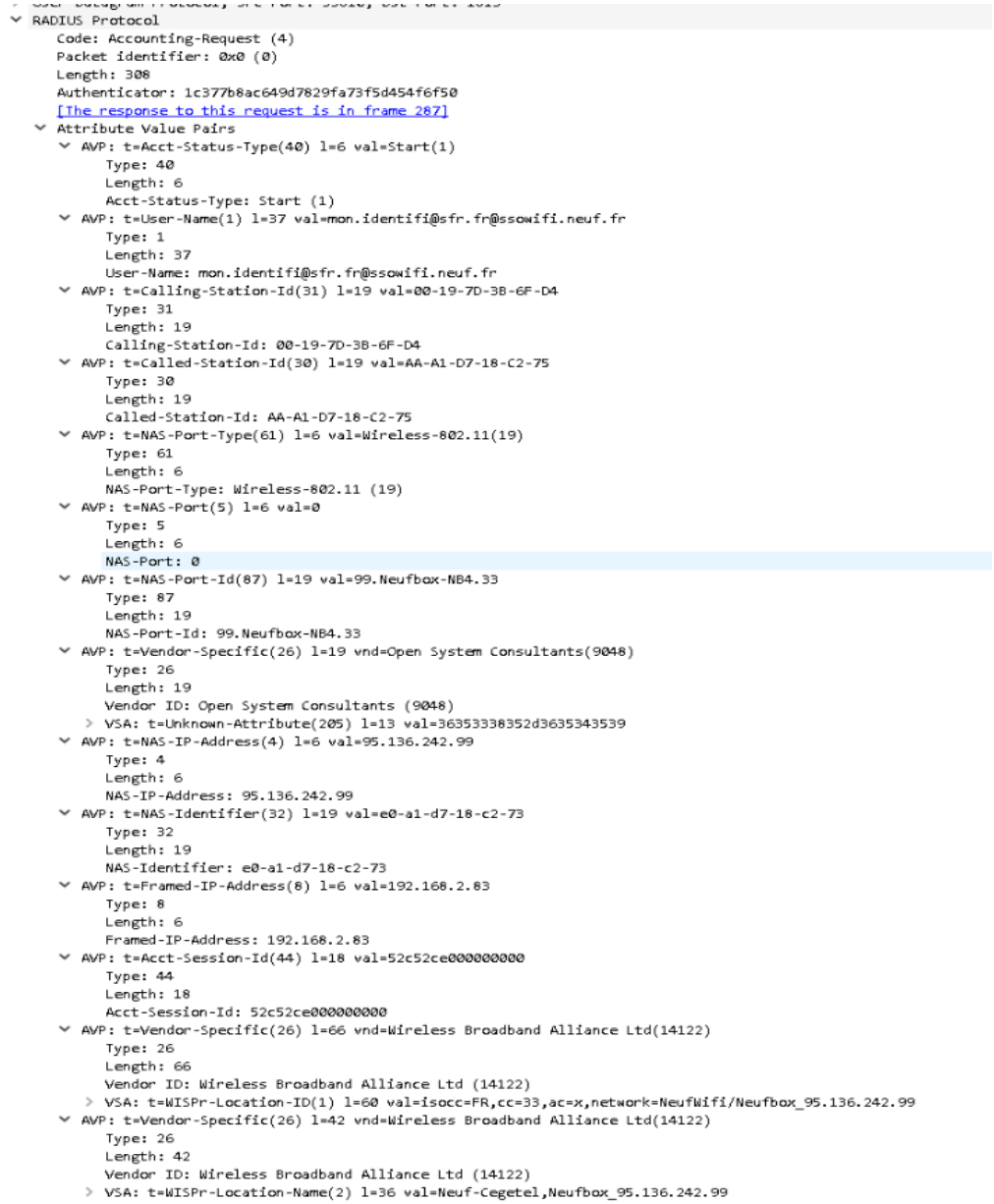


In the packet details pane, I've identified the Source port and Destination port in Packet 285.

The Source port is: e0:a1:d7:18:c2:72

The Destination port is: 80:fb:06:f0:45

In the packet details pane of Packet 285, I expanded the RADIUS Protocol line and delved deeper into the Attribute Value Pairs line. RADIUS, standing for Remote Authentication Dial-In User Service, serves as a crucial networking protocol responsible for Authentication, Authorization, and Accounting (AAA) management. It facilitates secure access to network resources, ensuring only authorized users gain entry. However, it's essential to recognize that within Packet 285, the RADIUS Attribute Value Pairs data contains highly sensitive information. This includes details such as the username, call station ID, connection type (Wireless), port ID (URL), OU name, internal IP schema, and wireless carrier. Such sensitive data underscores the importance of robust security measures to safeguard network integrity and protect user privacy.



Section 2, Part 2

Firstly, I launched the application from the vWorkstation taskbar. Once inside, I created a New Local Collection named "Jonathan_HotspotCapture_S2." After activating the new collection, I imported the HotspotCapture packet file previously explored in Part 1 of the lab. With the collection now open, I utilized the Investigator categories to pinpoint information regarding DNS sessions within the packet capture file. Lastly, I captured a screenshot showcasing essential details such as the Hostname Alias, Source IP Address, Destination IP Address, and source city fields of the perdu.com DNS session. This comprehensive approach aids in thorough analysis and understanding of network traffic patterns and potential security threats.

I've completed the steps outlined to navigate within NetWitness Investigator. Firstly, I launched the application from the vWorkstation taskbar. Once inside, I created a New Local Collection named "Jonathan_HotspotCapture_S2." After activating the new collection, I imported the HotspotCapture packet file previously explored in Part 1 of the lab. With the collection now open, I utilized the Investigator categories to pinpoint information regarding DNS sessions within the packet capture file.

The Hostname Alias associated with the perdu.com DNS session includes hotspot.wifi.sfr.fr, *.wifi.sfr.fr, perdu.com, ocsp.verisign.net, and ocsp.thawte.com. The Source IP Address ranges from 95.136.242.99 to 10.251.23.139, while the Destination IP Address spans across 109.0.74.75, 109.0.66.20, 80.118.192.115, 239.255.255.250, 208.97.177.124, 199.7.71.72, 109.6.1.72, and 62.39.3.142.



Assignment

1) Describe briefly what you learned or observed in the lab

In the process of investigating network traffic using both Wireshark and NetWitness Investigator, I navigated through various packet captures and collections to analyze DNS sessions, RADIUS protocols, and other network activities. In Wireshark, I meticulously examined packets, identifying source and destination MAC addresses, and source and destination ports, and delving into specific protocol details such as IEEE 802.11 QoS Data fields. The analysis revealed sensitive information, including usernames, connection types, and internal IP schemas, underscoring the importance of robust security measures.

Switching to NetWitness Investigator, I created a new local collection named "Jonathan_HotspotCapture_S2" and imported the HotspotCapture packet file for further analysis. Within the collection, I utilized Investigator categories to explore DNS sessions and identified key information such as Hostname Alias, Source IP Address, Destination IP Address, and source city fields. Notably, the DNS sessions revealed various aliases for the perdu.com domain, including hotspot.wifi.sfr.fr and ocsp.verisign.net, reflecting a diverse range of network activities.

The investigation uncovered a wide array of IP addresses involved in the communication, spanning across different destinations and origins. From IP addresses in France to others around the world, the network traffic depicted a complex web of interactions, highlighting the global nature of modern network communications.

Overall, the thorough analysis conducted using both Wireshark and NetWitness Investigator provided valuable insights into network traffic patterns, potential security threats, and the intricacies of data transmission. This comprehensive approach facilitates effective network monitoring and troubleshooting, enabling organizations to maintain robust cybersecurity measures and ensure the integrity and security of their networks.

2) Explain the benefits of encryption using a public-private key pair and the concept of non-repudiation in relation to a digital signature (email instructions updated).

Public-key encryption is a cornerstone of modern cryptography, providing a robust framework for secure communication, data integrity, and authentication in digital environments. By leveraging asymmetric cryptographic algorithms, public-key encryption allows entities to encrypt and decrypt data using separate keys, known as the public key and private key, respectively. This approach offers numerous advantages, including flexible key management, scalability, enhanced trustworthiness, legal recognition, and compliance with regulatory standards. In this discussion, we will explore the ramifications of encrypting files using a public key in detail, delving into the intricacies of public-key encryption issues and their implications for data security and integrity. Through a comprehensive analysis of key concepts such as non-repudiation, trustworthiness, and legal compliance, we will gain a deep understanding of the practical applications and challenges associated with public-key encryption in today's digital landscape.

Flexibility in Key Management: Public-key encryption allows for flexible key management and distribution. Each entity generates its own unique public-private key pair, eliminating the need for a shared secret key. This decentralized approach simplifies key management and enhances security by preventing the compromise of one party's key from affecting others.

Scalability Across Networks: Public-key encryption scales effectively in large networks with numerous users and devices. Each entity can have its unique public-private key pair, enabling secure communication with any other entity in the network without requiring pre-shared keys. This scalability makes public-key encryption suitable for a wide range of applications, from secure email communication to online transactions.

Enhanced Trustworthiness: Digital signatures provide a high level of trust in digital transactions. By signing a message with their private key, Party A creates a digital signature that includes cryptographic hash functions representing the message's content. Any alteration to the message would result in a different hash value, rendering the signature invalid. Therefore, the recipient can trust both the message's integrity and the signer's authenticity.

Legal Recognition and Compliance: Digital signatures hold legal recognition and compliance in many jurisdictions globally. Legislation such as the Electronic Signatures in Global and National Commerce Act (ESIGN) in the United States and the eIDAS Regulation in the European Union establish the legal validity of digital signatures for electronic transactions. This legal framework ensures that digitally signed documents carry the same legal weight as their paper counterparts, streamlining business processes and reducing paperwork.

Non-Repudiation: Non-repudiation is a critical aspect of digital signatures. It ensures that the signer cannot later deny their involvement in creating the signed message. When Party A signs a message with their private key, the digital signature serves as evidence of Party A's consent and intention to be bound by the contents of the message. Since only Party A possesses the private key, they cannot repudiate their signature later, providing legal and evidentiary value to the signed document.

Comprehensive Audit Trails: Digital signatures offer comprehensive audit trails of digital transactions. Each signed message includes metadata such as the signer's identity, timestamp, and cryptographic hash of the signed data. These metadata elements facilitate tracking and verifying the authenticity, integrity, and non-repudiation of digital transactions, promoting accountability and compliance with regulatory standards.

Cross-Platform Compatibility: Public-key encryption and digital signatures are platform-independent, making them compatible across various operating systems, devices, and applications. This compatibility ensures interoperability and seamless integration into diverse IT environments, empowering organizations to leverage encryption and digital signature technologies across their entire ecosystem.

Encryption using a public-private key pair and digital signatures offers numerous benefits, including flexible key management, scalability, enhanced trustworthiness, legal recognition and compliance, comprehensive audit trails, non-repudiation, and cross-platform compatibility. These

advantages make them essential tools for securing digital communications, transactions, and documents in today's interconnected and digitized world. The detailed explanation provided demonstrates a deep understanding of public key encryption issues and their ramifications.

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